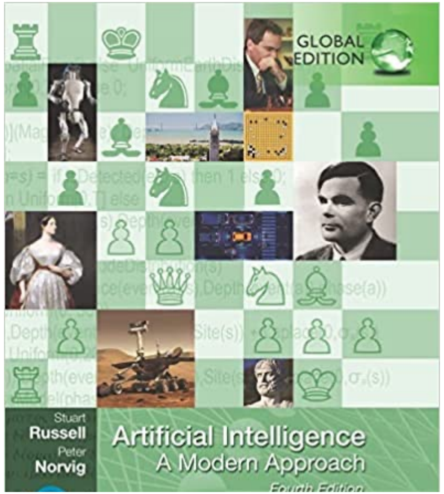
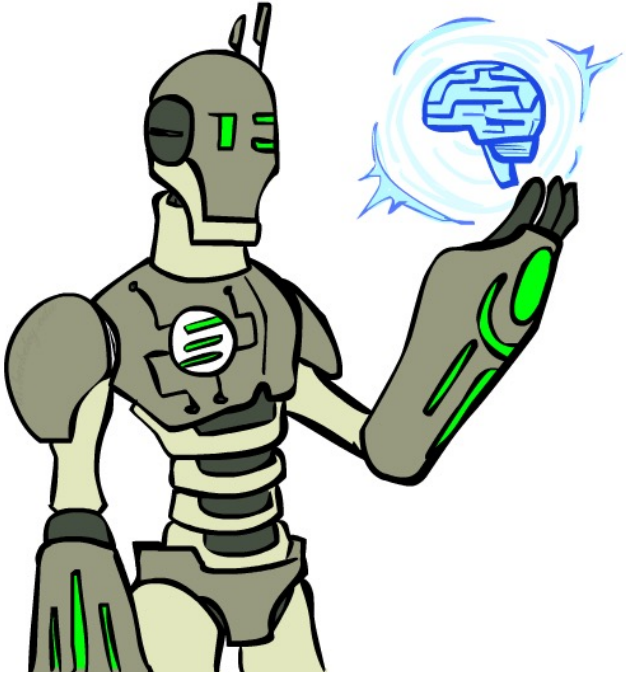
INTRODUCTION



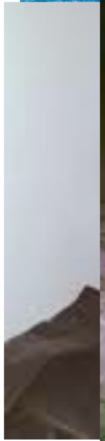
Slides by Stuart Russell

Today

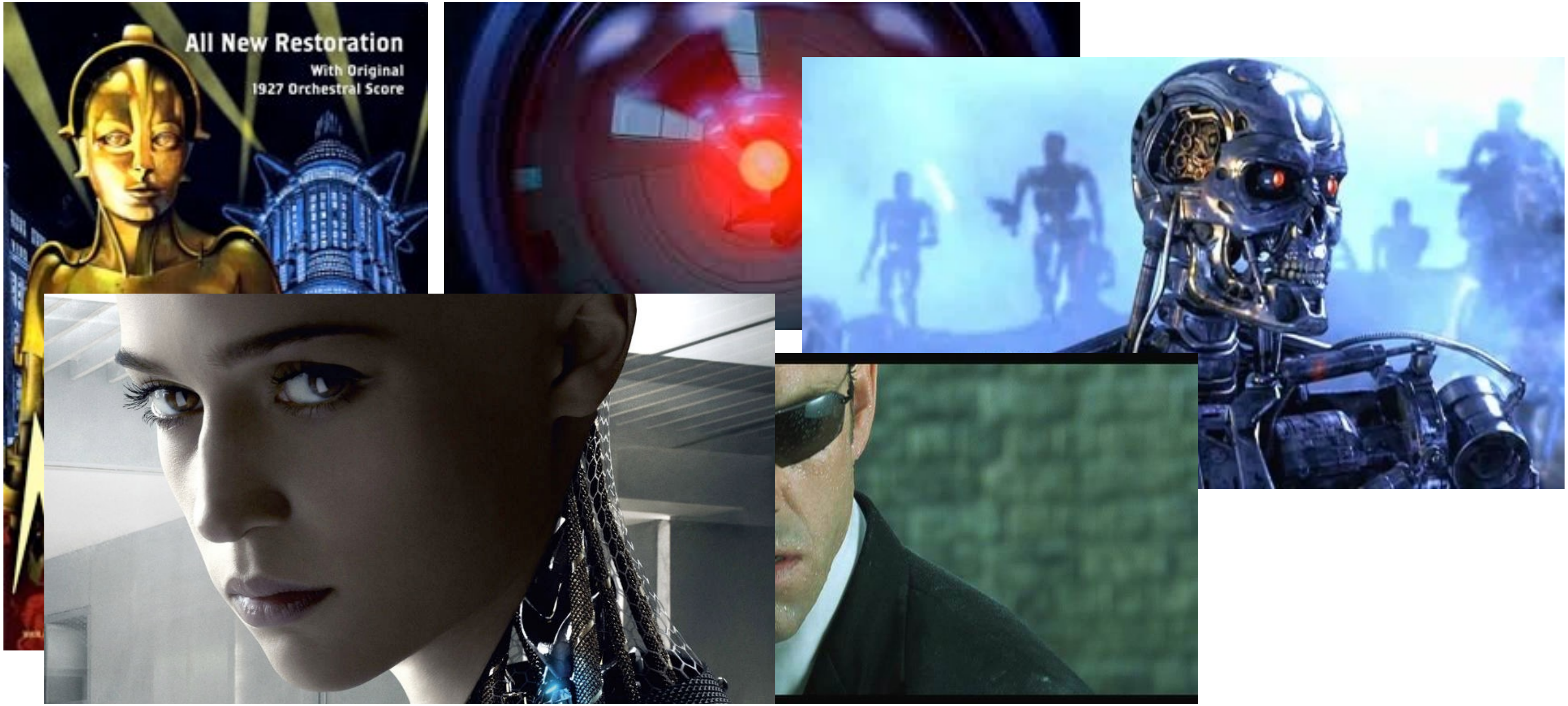
- What is artificial intelligence?
- Where are we and how did we get here?
- How do we think about the design of AI systems?



Movie AI

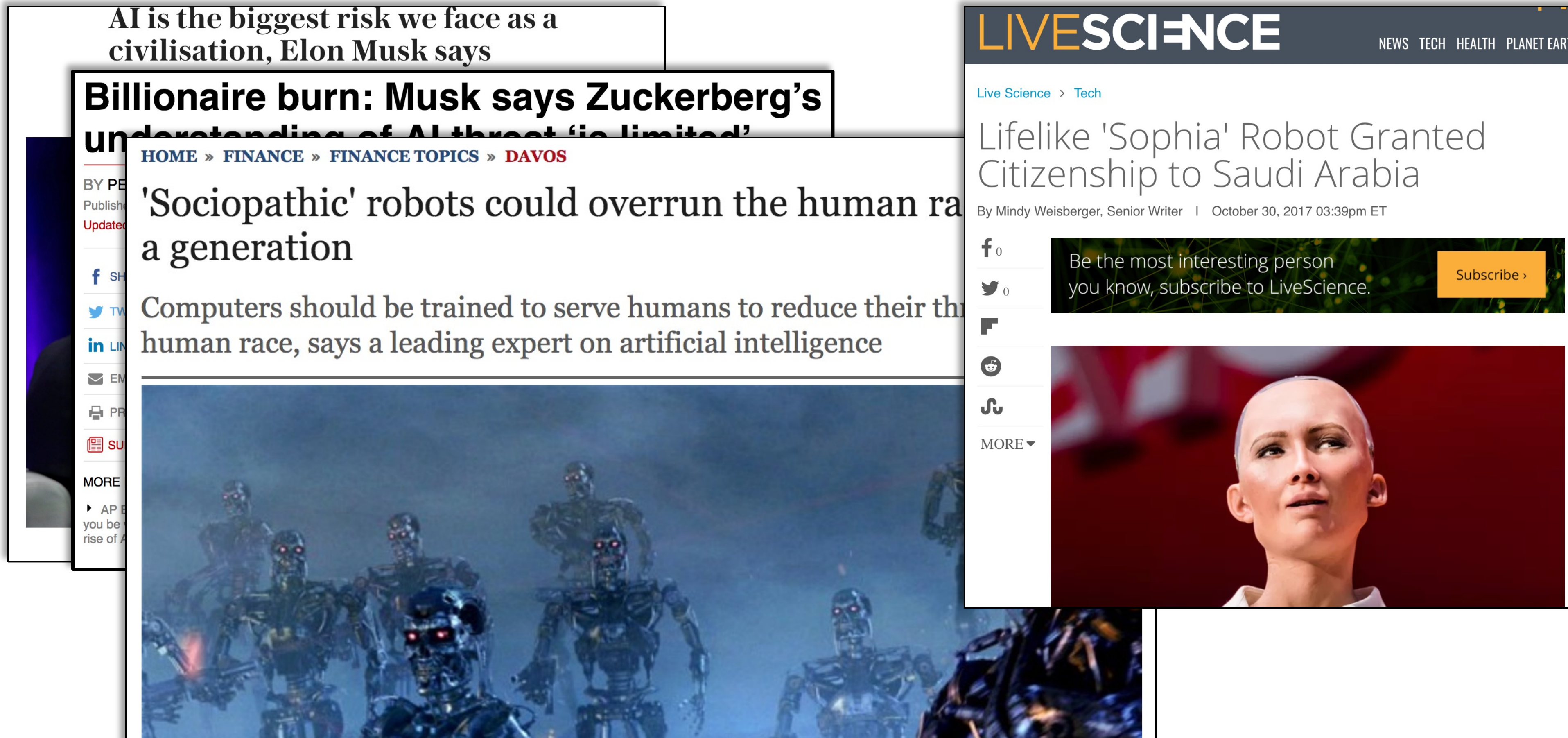


Movie AI




YESTERDAY DR. WILL CASTER WAS ONLY HUMAN
JOHNNY DEPP
REBECCA HALL
PAUL BETTANY
KATE MARA
CILLIAN MURPHY
AND MORGAN FREEMAN
TRANSCENDENCE
entertainmentfilms.co.uk

News AI


AI is the biggest risk we face as a civilisation, Elon Musk says
Billionaire burn: Musk says Zuckerberg's
HOME » FINANCE » FINANCE TOPICS » DAVOS
'Sociopathic' robots could overrun the human ra
a generation
Computers should be trained to serve humans to reduce their th
human race, says a leading expert on artificial intelligence
LIVESCIENCE
NEWS TECH HEALTH
Live Science › Tech
Lifelike 'Sophia' Robot Granted Citizenship to Saudi Arabia
By Mindy Weisberger, Senior Writer | October 30, 2017 03:39pm ET
Be the most interesting person you know, subscribe to LiveScience.
Subscribe ›
MORE

News AI

TECH • ARTIFICIAL INTELLIGENCE

United Kingdom Plans $1.3 Billion Artificial Intelligence Push

France to spend $1.8 billion on AI to compete with U.S., China

EU wants to invest £18b

development

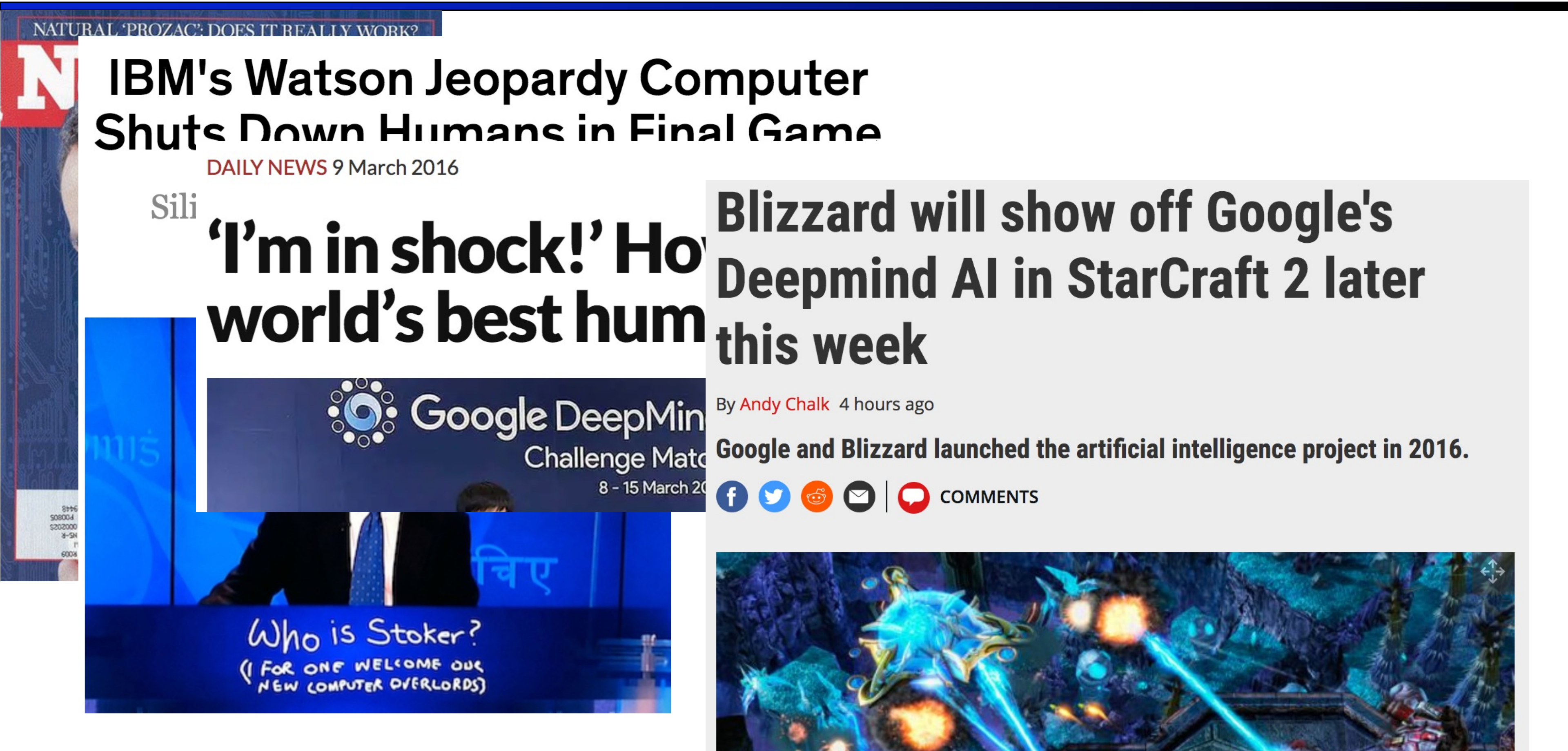
China's Got a Huge Art

Intelligence Plan

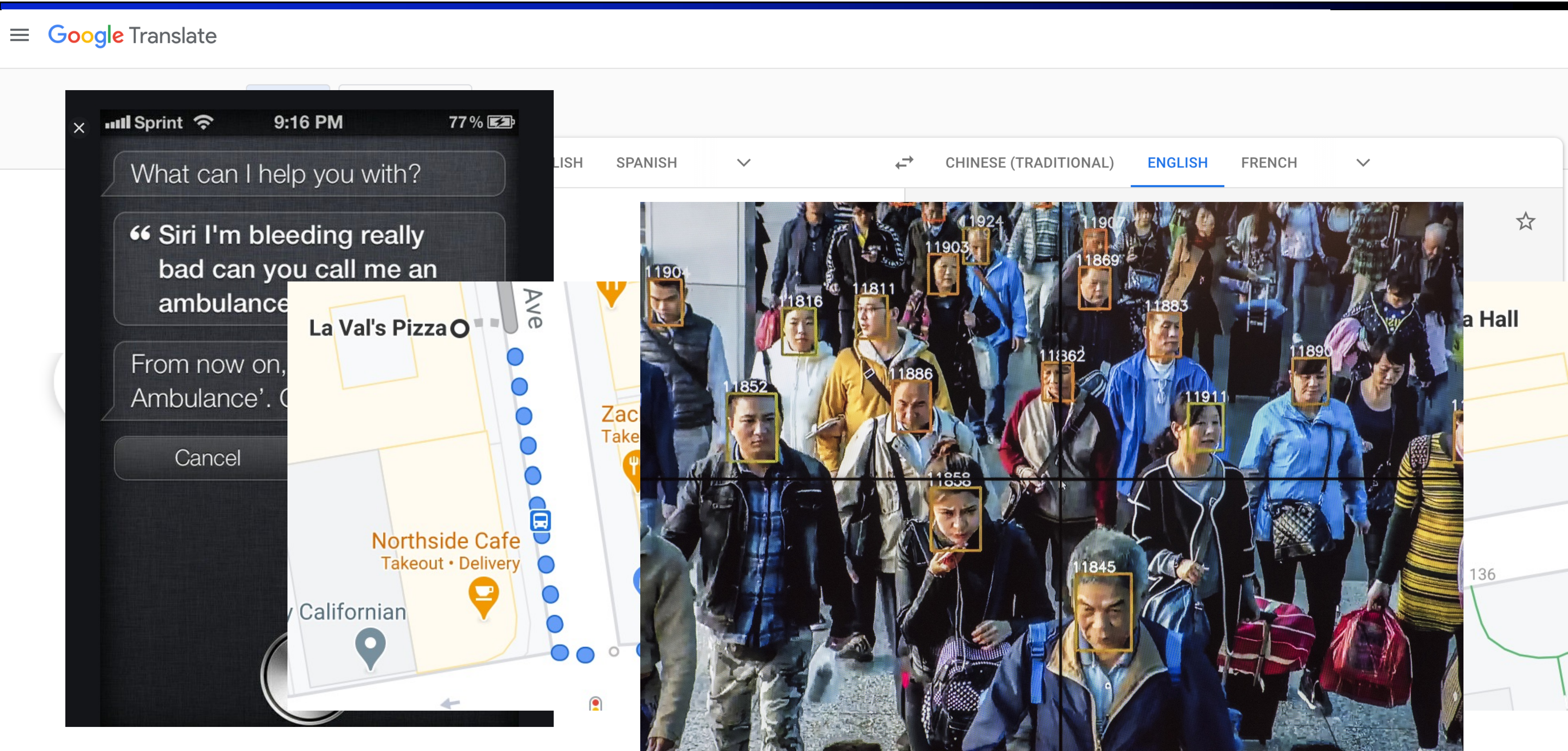
'Whoever leads in AI will rule the world': Putin to Russian children on Knowledge Day

Published time: 1 Sep, 2017 14:08
Edited time: 1 Sep, 2017 14:40

News AI


NATURAL 'PROZAC': DOES IT REALLY WORK?
IBM's Watson Jeopardy Computer Shuts Down Humans in Final Game
DAILY NEWS 9 March 2016
'I'm in shock!' Ho
world's best hum
Google DeepMin
Challenge Matc
8 - 15 March 20
Who is Stoker?
(I FOR ONE WELCOME OUR NEW COMPUTER OVERLORDS)
Blizzard will show off Google's Deepmind AI in StarCraft 2 later this week
By Andy Chalk 4 hours ago
Google and Blizzard launched the artificial intelligence project in 2016.
COMMENTS

Real AI

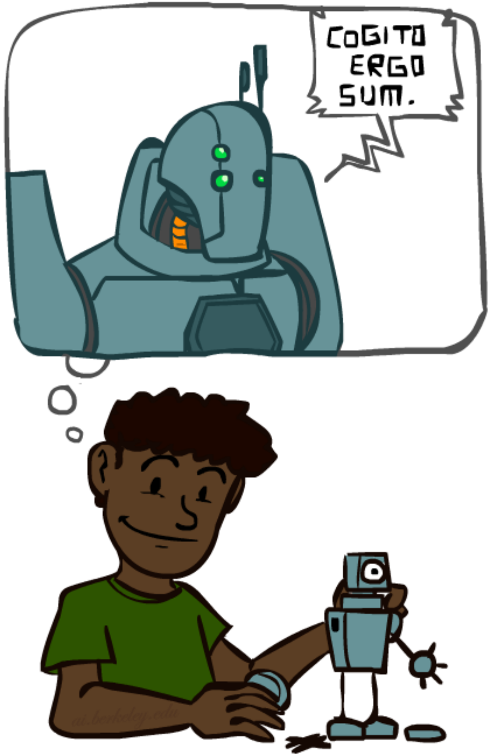


Google Translate
LISH
SPANISH
CHINESE (TRADITIONAL)
ENGLISH
FRENCH
Sprint
9:16 PM
77 %
What can I help you with?
“ Siri I'm bleeding really bad can you call me an ambulance
From now on,
Ambulance'.
Cancel
La Val's Pizza
Ave
Northside Cafe
Takeout • Delivery
Californian
a Hall
136


TUG
CAUTION
MAY CONTAIN
CHEMOTHERAPY DRUG
CAUTION
MAY CONTAIN
CHEMOTHERAPY DRUG


Boston Dynamics

A (Short) History of AI



Demo: HISTORY – MT1950.wmv

Origins of AI

- Influences:
  - **Philosophy** (reasoning, planning, learning, science, automation)
  - **Mathematics** (logic, probability, optimization)

> Aristotle: For if every instrument could accomplish its own work, obeying or anticipating the will of others . . . if, in like manner, the shuttle would weave and the plectrum touch the lyre without a hand to guide them, chief workmen would not want servants, nor masters slaves

  - **Linguistics** (grammars, formal representation of meaning)

- Prehistory:
  - Leibniz “Calculemus!” (early 1700)
  - Babbage Difference Engine (and Analytical Engine (first general-purpose computer, mechanical)
  - Lovelace: “a thinking machine” for “all subjects in the universe.”

Prehistory of AI

1931

The Austrian Kurt Goedel shows that in first-order predicate logic all true statements are derivable. In higher order logics, on the other hand, there are true statements that are unprovable.

1937

Alan Turing points out the limits of intelligent machines with the halting problem.

1943

McCulloch and Pitts model neural networks and make the connection to propositional logic.

1950

Alan Turing defines machine intelligence with the Turing test and writes about learning machines and genetic algorithms

1951

Marvin Minsky develops a neural network machine. With 3000 vacuum tubes he simulates 40 neurons

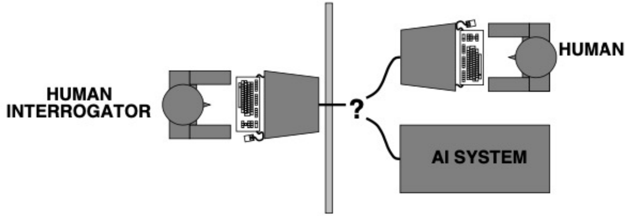
Added by Giuseppe De Giacomo

Turing Test

Turing (1950) "Computing machinery and intelligence":

◊ "Can machines think?" ⟶ "Can machines behave intelligently?"

◊ Operational test for intelligent behavior: the Imitation Game



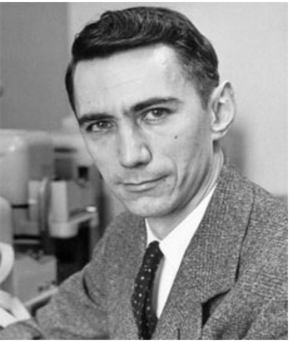
◊ Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes

◊ Anticipated all major arguments against AI in following 50 years

◊ Suggested major components of AI: knowledge, reasoning, language understanding, learning

Problem: Turing test is not reproducible, constructive, or amenable to mathematical analysis

AI's official birth: Dartmouth, 1956



"An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ***We think that a significant advance can be made if we work on it together for a summer."***

John McCarthy and Claude Shannon
Dartmouth Workshop Proposal

Shakey the Robot (1966-1972)

- Short video on Shakey
  - https://www.youtube.com/watch?v=7bsEN8mwUB8

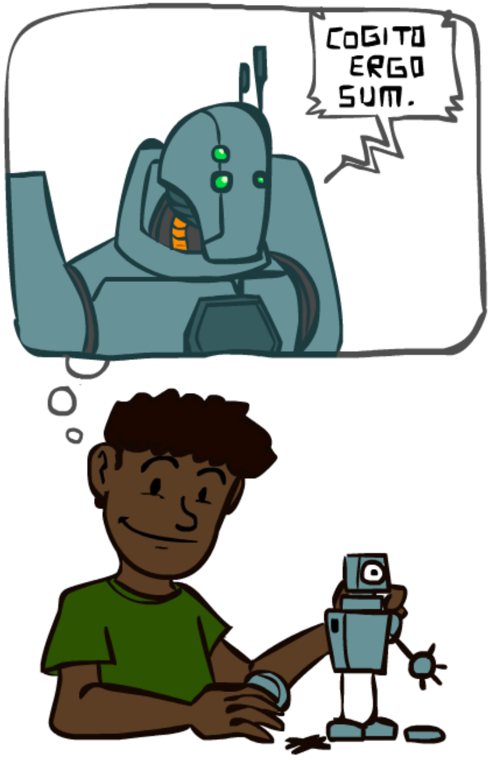
- Long video on Shakey
  - https://www.youtube.com/watch?v=GmU7SimFkpU

Wiki: https://en.wikipedia.org/wiki/Shakey_the_robot

Added by Giuseppe De Giacomo

A (Short) History of AI

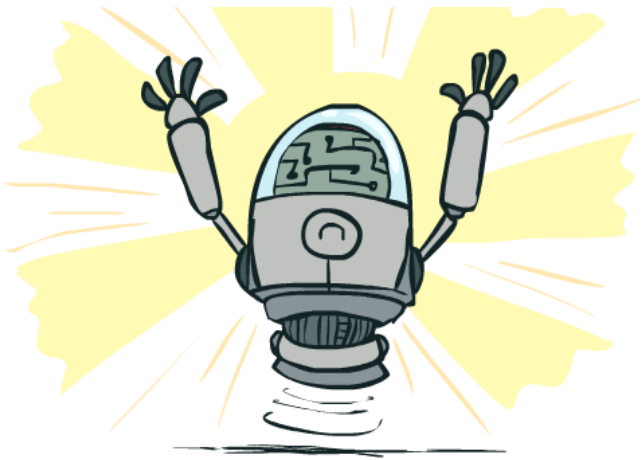
- 1940-1950: Early days
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: **Turing's "Computing Machinery and Intelligence"**
- 1950—70: Excitement: Look, Ma, no hands!
  - 1950s: Early AI programs: chess, checkers (RL), theorem proving
  - 1956: **Dartmouth meeting: "Artificial Intelligence" adopted**
  - 1965: Robinson's complete algorithm for logical reasoning
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1988—93: Expert systems industry busts: "AI Winter"
- 1990— 2012: Statistical approaches + subfield expertise
  - Resurgence of probability, focus on uncertainty
  - **General increase in technical depth**
  - Agents and learning systems… "AI Spring"?
- 2012— ___: Excitement: Look, Ma, no hands again?
  - Big data, big compute, deep learning
  - **AI used in many industries**



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Play a decent game of table tennis?
- ✓ Play a decent game of Jeopardy?
- ✓ Drive safely along a curving mountain road?
- ✗ Drive safely along Telegraph Avenue?
- ✓ Buy a week's worth of groceries on the web?
- ✗ Buy a week's worth of groceries at Berkeley Bowl?
- ? Discover and prove a new mathematical theorem?
- ✗ Converse successfully with another person for an hour?
- ? Perform a surgical operation?
- ✓ Translate spoken Chinese into spoken English in real time?
- ? Fold the laundry and put away the dishes?
- ✗ Write an intentionally funny story?



Unintentionally Funny Stories

Once upon a tir
and a vain crow
in his tree, hold
mouth. He not
piece of cheese
swallowed the
the crow. The

Janelle Shane
@JanelleCShane

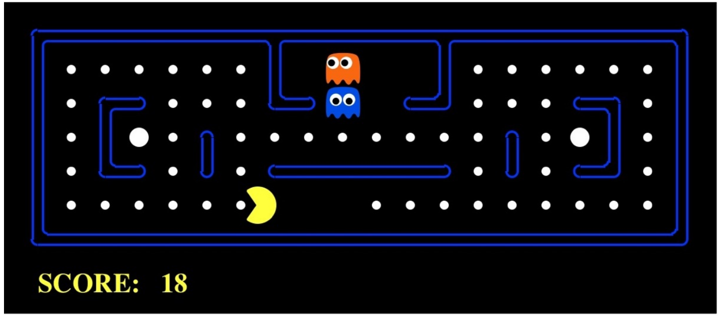
Follow

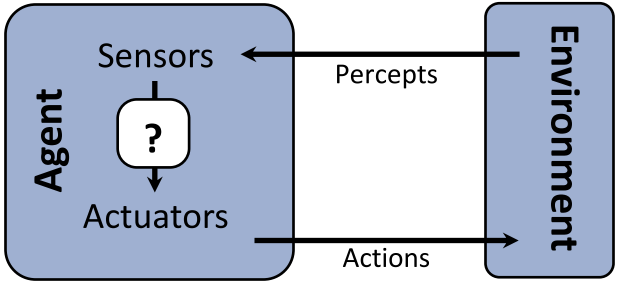
Tried retraining the neural net on just "what do you get when you cross a X with a X?" jokes. Results did not improve. And for some reason, bungees are its favorite thing.

What do you get when you cross a dog and a vampire?
A bungee
What do you get when you cross a cow with a rhino?
A bungee with a dog
What do you get when you cross a street and a cow?
A bungee with a bungee and a rhino
What do you get when you cross a pig with a cow with a party?
Because the engineers with a dog

AI as Designing Rational Agents

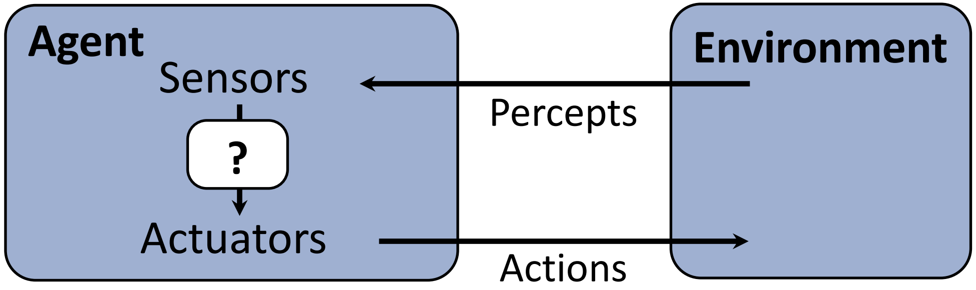
- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its expected **utility**.
- Characteristics of the **sensors, actuators, and environment** dictate techniques for selecting rational actions
- **This course** is about:
  - General AI techniques for many problem types
  - Learning to choose and apply the technique appropriate for each problem





Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes

Agents and environments



- An agent ***perceives*** its environment through ***sensors*** and ***acts*** upon it through ***actuators*** (or *effectors*, depending on whom you ask)
- The ***agent function*** maps percept sequences to actions
- It is generated by an ***agent program*** running on a ***machine***

Rational agents

An agent is an entity that perceives and acts

This course is about designing rational agents

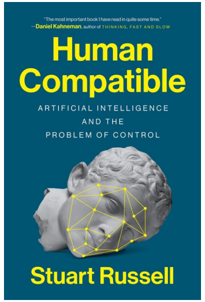
Abstractly, an agent is a function from percept histories to actions:

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$

For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

Caveat:

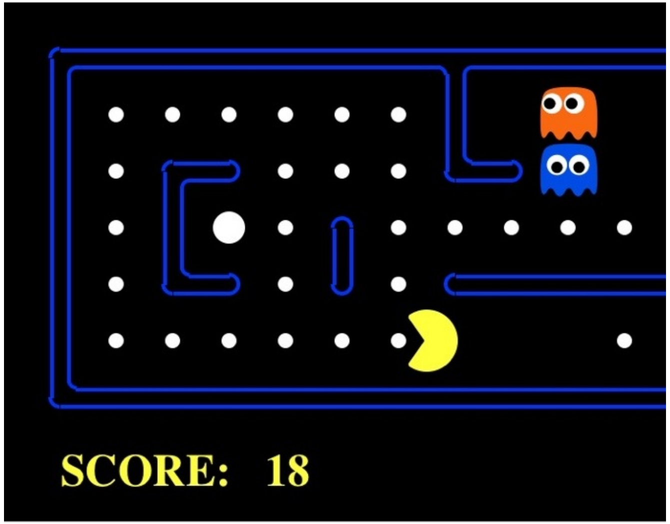
1. Computational **limitations** make perfect rationality unachievable
   → design best program for given resources
2. Herbert Simon's **satisficing behavior**: how to apply it to AI?
3. Stuart Russell: be careful **not to act as a fanatic**!



A human agent in Pacman

The task environment - PEAS

- Performance measure
  - -1 per step; + 10 food; +500 win; -500 die; +200 hit scared ghost
- Environment
  - Pacman dynamics (incl ghost behavior)
- Actuators
  - Left Right Up Down or NSEW
- Sensors
  - Entire state is visible (except power pellet duration)



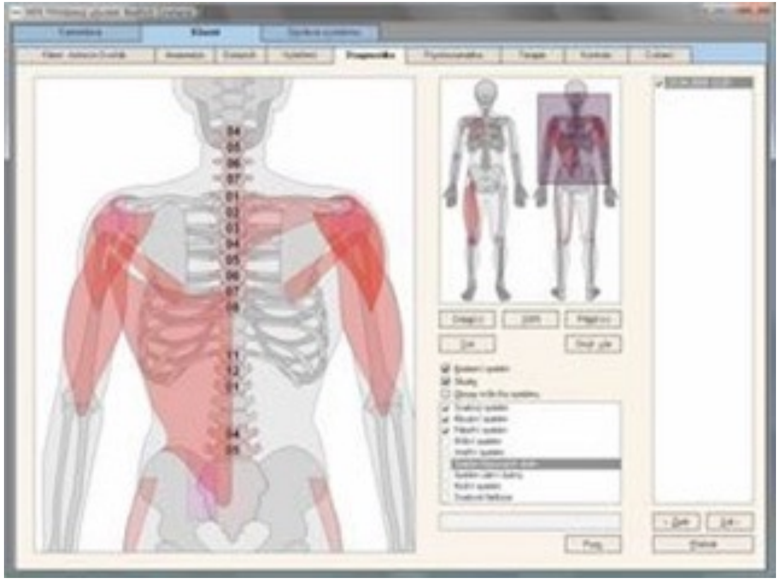
PEAS: Automated taxi

- Performance measure
  - Income, happy customer, vehicle costs, fines, insurance premiums
- Environment
  - US streets, other drivers, customers, weather, police…
- Actuators
  - Steering, brake, gas, display/speaker
- Sensors
  - Camera, radar, accelerometer, engine sensors, microphone, GPS

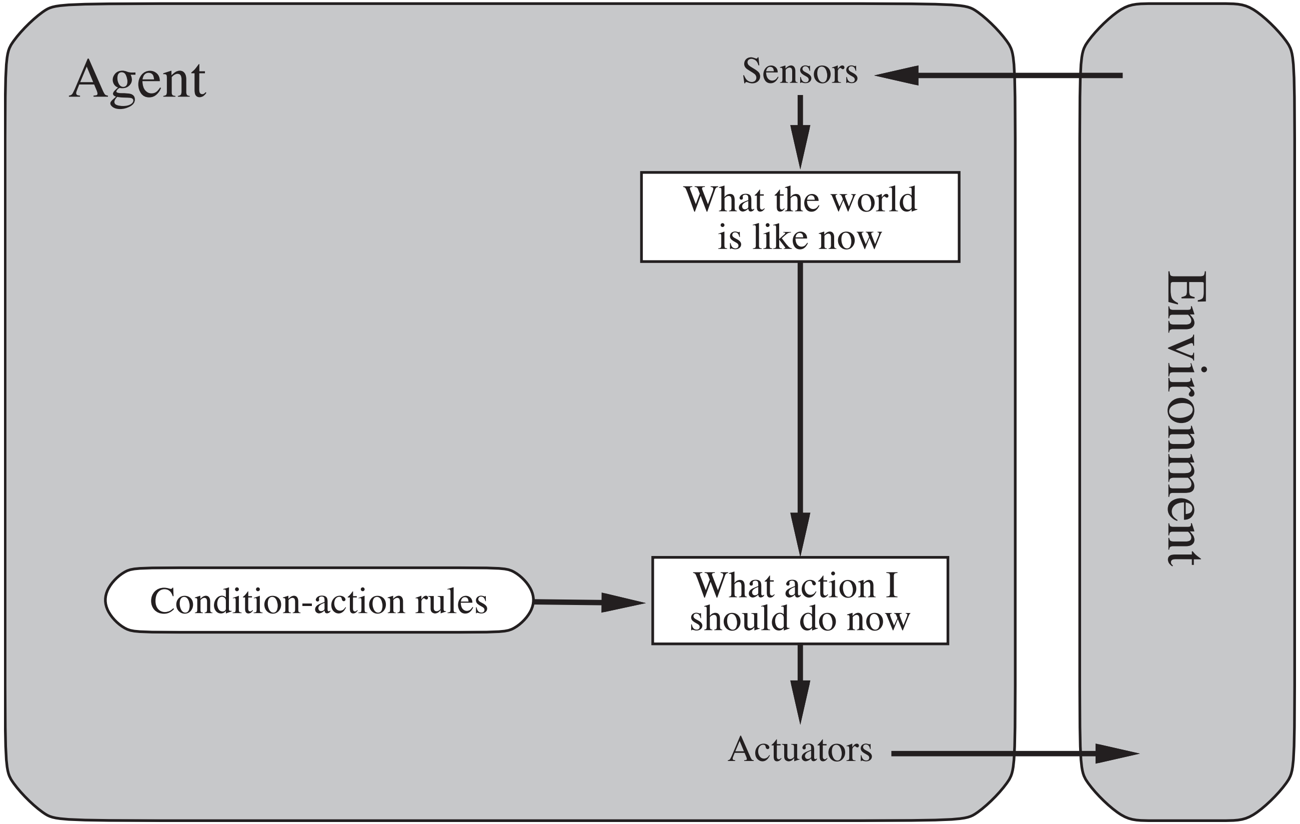
Image: http://nypost.com/2014/06/21/how-google-might-put-taxi-drivers-out-of-business/

PEAS: Medical diagnosis system

- Performance measure
  - Patient health, cost, reputation
- Environment
  - Patients, medical staff, insurers, courts
- Actuators
  - Screen display, email
- Sensors
  - Keyboard/mouse



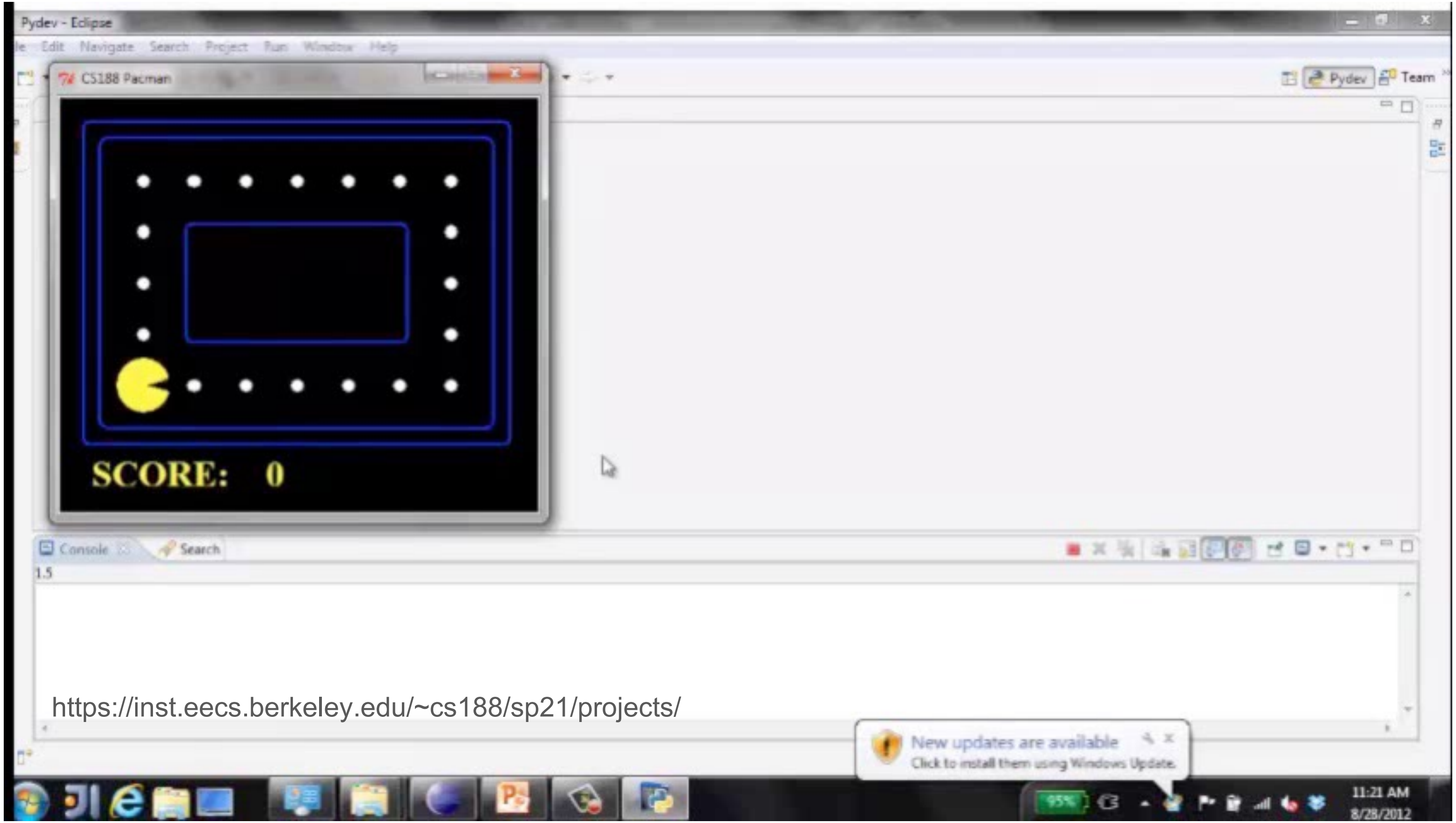
Simple reflex agents



Pacman ***agent program*** in Python

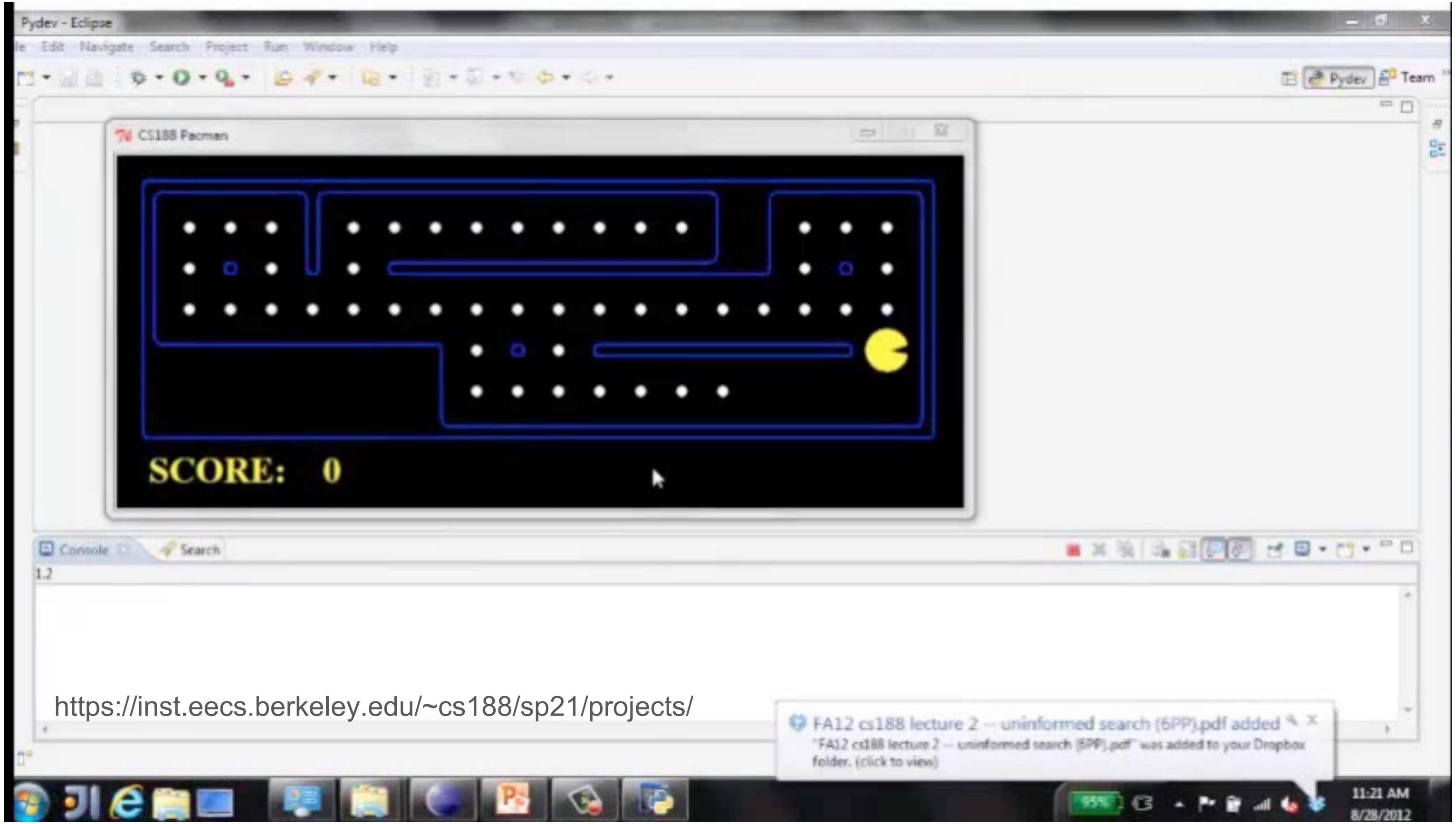
```
class GoWestAgent(Agent):

    def getAction(self, percept):
        if Directions.WEST in percept.getLegalPacmanActions():
            return Directions.WEST
        else:
            return Directions.STOP
```

Eat adjacent dot, if any



https://inst.eecs.berkeley.edu/~cs188/sp21/projects/

Eat adjacent dot, if any

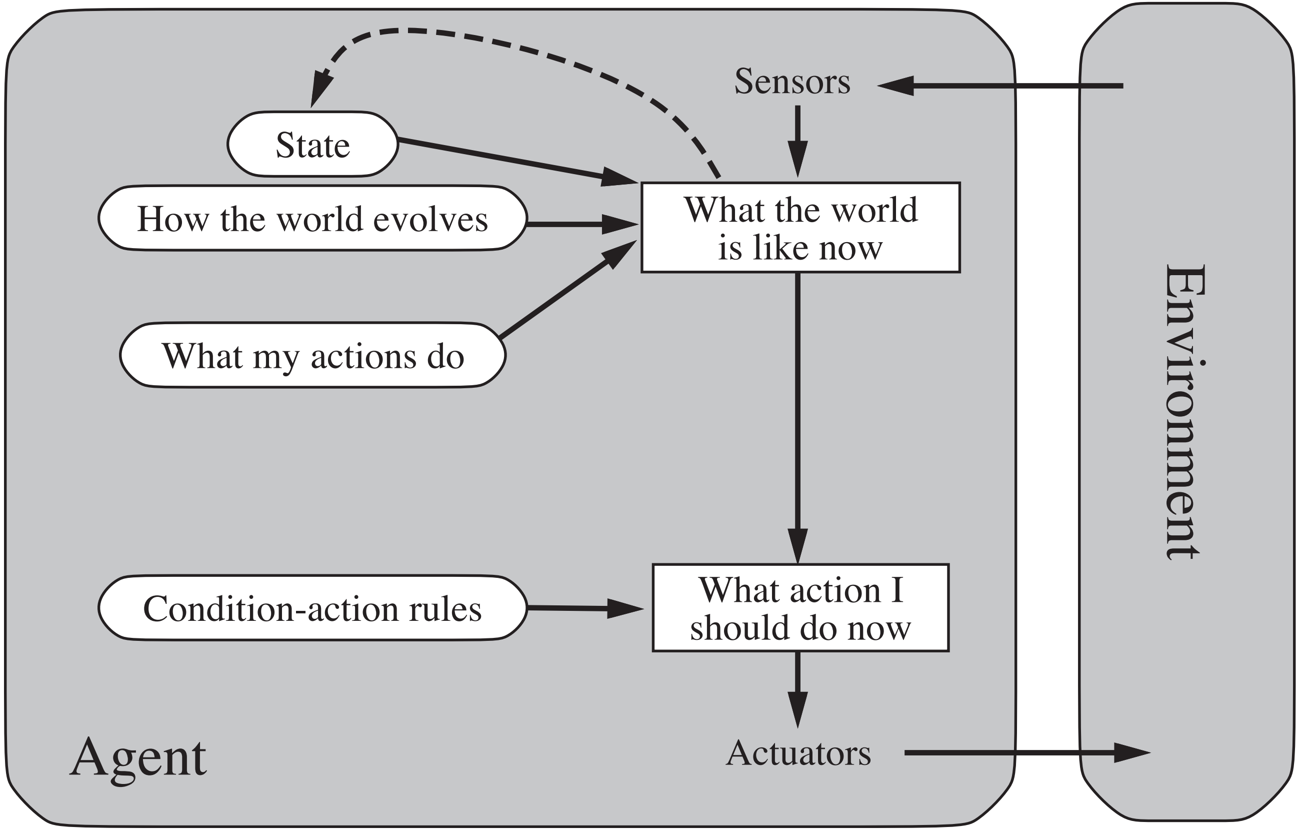


https://inst.eecs.berkeley.edu/~cs188/sp21/projects/

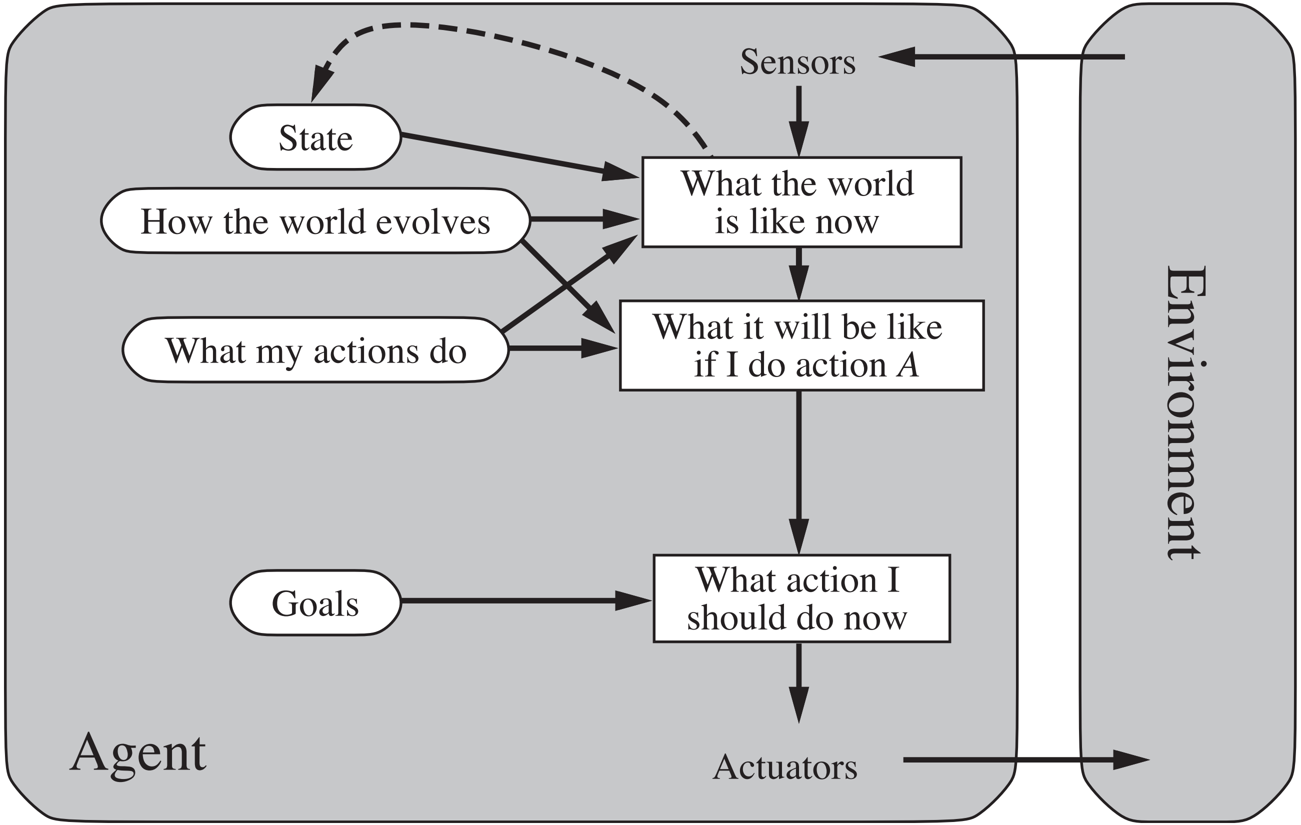
Pacman agent contd.

- Can we (in principle) extend this reflex agent to behave well in all standard Pacman environments?
  - No – Pacman is not quite fully observable (power pellet duration)
  - Otherwise, yes – we can (*in principle*) make a lookup table…..
  - *How large would it be?*

Reflex agents with state



Goal-based agents



What is AI?

EU vision:

- Artificial intelligence (AI) refers to systems that display intelligent behavior by analyzing their environment and taking actions with some degree of autonomy to achieve specific goals.

- AI-based systems can be purely software-based, acting in the virtual world (e.g., voice assistants, image analysis software, search engines, speech and face recognition systems) or AI can be embedded in hardware devices (e.g., advanced robots, autonomous cars, drones or Internet of Things applications).

Added by Giuseppe De Giacomo
Courtesy of Daniele Nardi

Course Description

AI&ML introduces the basics of Artificial Intelligence and Machine Learning.

- This course focuses on transferring knowledge and capability to the "machine" (the intelligent agent) to make it able to understanding the world and answering questions about it, as well as to acting on it to accomplish desired tasks.

- The course will present the basic techniques developed in the field of Artificial Intelligence, concerning modeling the world and exploiting the model for reasoning and planning, focusing on discrete models.

- It will also present basic concepts and techniques from Machine Learning for learning from real world data or from simulations, how to answer and how to act in the world with the purpose of achieving desired goals.

Teaching material:

- [1] Course slides 2022/2023 and additional material.
- [2] Artificial Intelligence: A Modern Approach, Global Edition, 4th Edition by Stuart Russell, Peter Norvig, Pearson 2020 (selected chapters)
- [3] Reinforcement Learning: An Introduction, 2nd Edition by Richard S. Sutton and Andrew G. Barto. MIT Press, Cambridge, MA, 2018. (Freely available online)

Course Contents

Artificial Intelligence (6CFU)

- Introduction to Artificial Intelligence
- Knowledge Representation
- Propositional Logic
  - SAT
  - Tableaux
- First-Order Logic
  - Reasoning in First-Order Logic
  - Tableaux
- Reasoning about Actions
- Deterministic Planning Domains
  - Planning Domain Description Language (PDDL)
  - Planning in Deterministic Planning Domains
  - Search, Heuristics, Best-first, A*
- Nondeterministic Planning Domains (FOND)
  - Planning in Nondeterministic Planning Domains
  - Game Theoretic View
  - Adversarial Search, Search in AND-OR Graphs
- Generalized Forms of Planning
  - Partial Observability
  - Temporally Extended Goals
  - Program Synthesis (Formal Methods)

Machine Learning (3CFU)

- Introduction to Machine Learning
- Supervised Learning
- Deep Learning
- Reinforcement Learning